Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A – Signal and Image Processing

Lecture 12 - Applications of signal and image processing in visual AI



Slides inspired from
Stanford course EE269 Signal Processing for Machine Learning by Mert Pilanci
Efficient Image Processing with Deep Neural Networks, Tutorial ICIP 2019 by Vivienne Sze and Tien-Ju Yang

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Definition of Machine Learning

Definition of Machine Learning

- Samuel (1959): Field of study that gives computers the ability to learn without being explicitly programmed.
- Kevin Murphy (2012): Algorithms that automatically detect patterns in data use the uncovered patterns to predict future data or other outcomes of interest

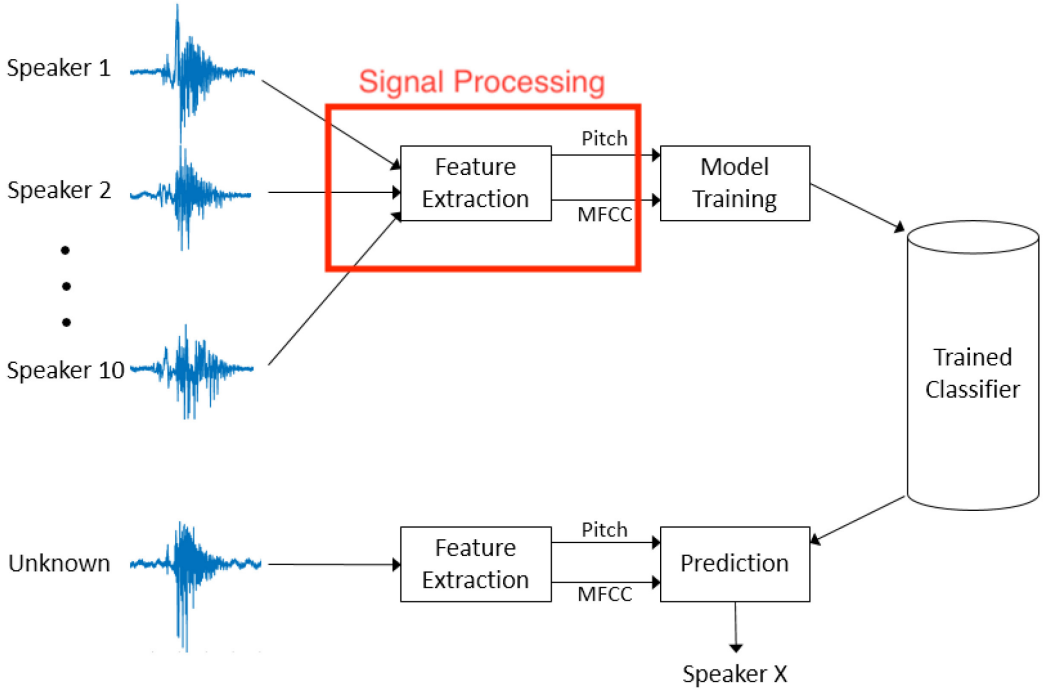
When to apply Machine Learning

- Humans are unable to explain their expertise (e.g. Speech recognition, vision, language)
- Solution changes with time (e.g. tracking, noise cancellation, adaptive filtering)
- Solution needs to be adapted to particular cases (e.g. biometrics, personalization)

Examples

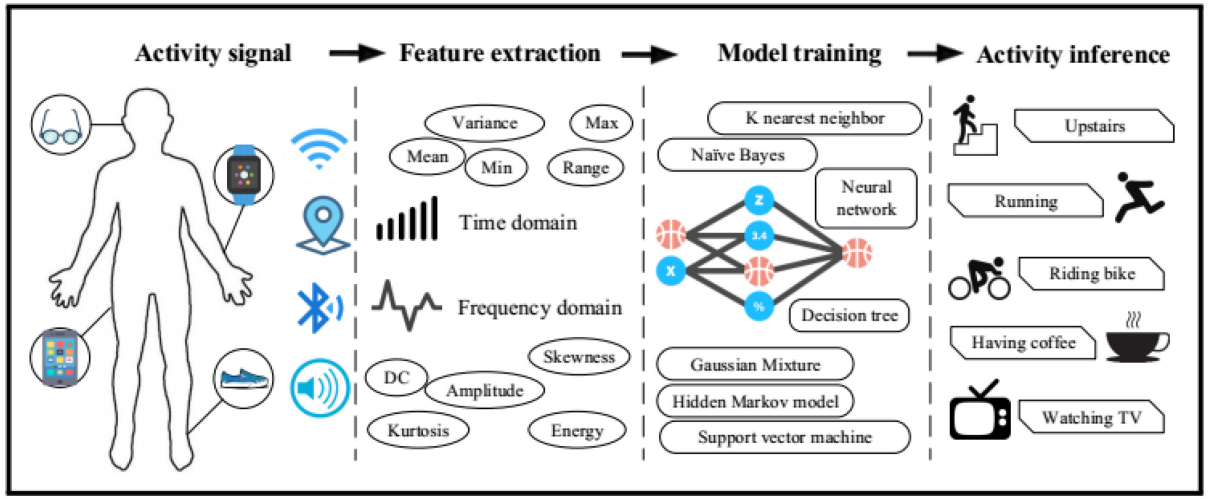
Speaker Identification



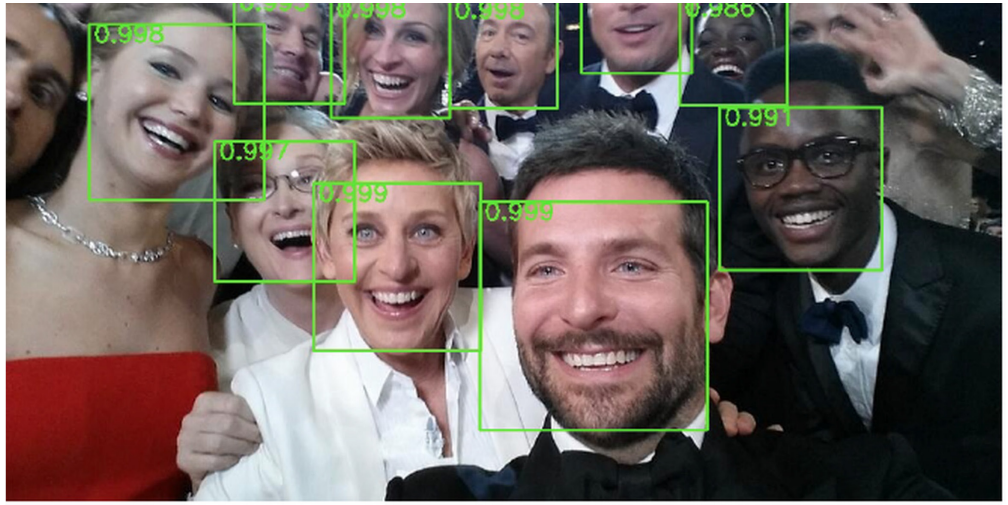


Examples

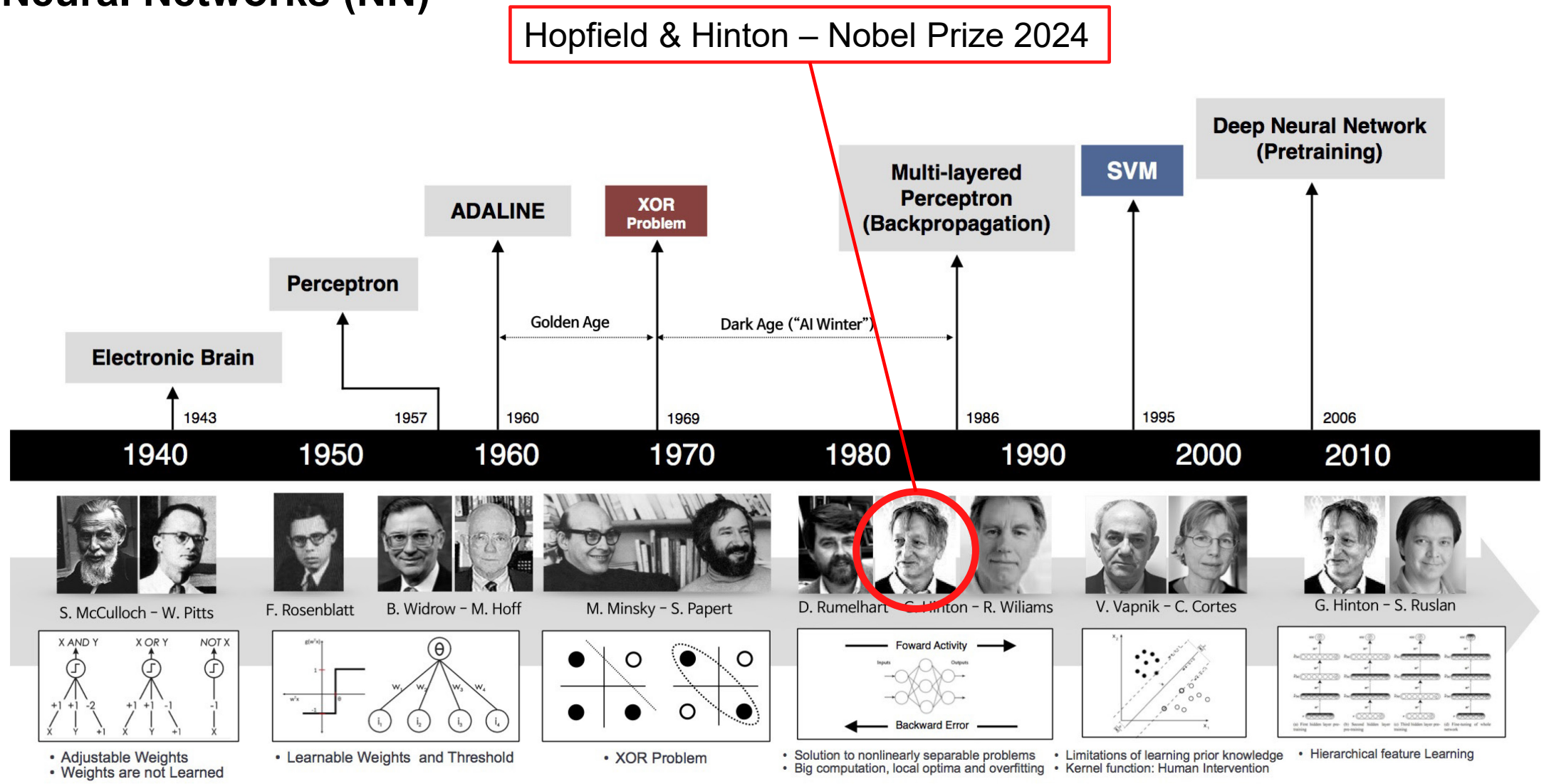
Activity recognition



Face detection

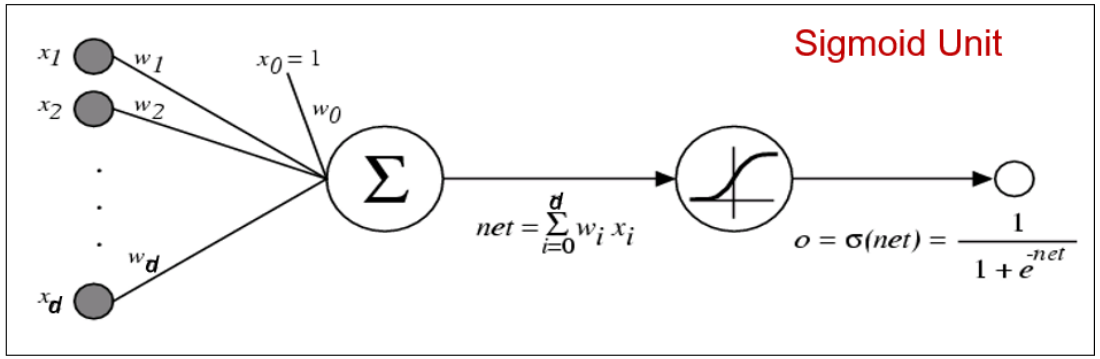


Neural Networks (NN)



Neural networks (NN)

- Neuron - perceptron



Neuron/Node: Basic unit of a neural network

It comprises of:
Weight: A connection between two neuron with value associated with it.

net : All the weighted inputs are added together using:

$$net = \sum_{i=0}^{d} w_i x_i$$

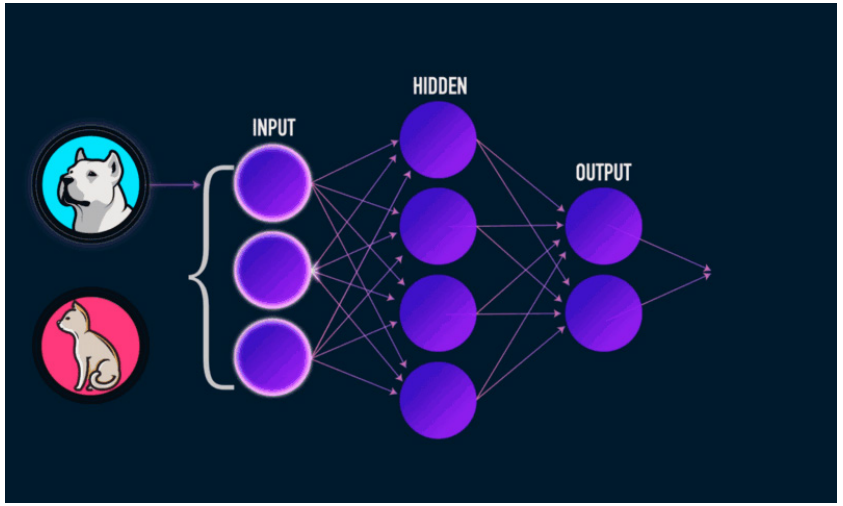
Where d is total number of connections.

Activation functions: Used to introduce non-linearity in neural networks (defines the output of each node).

Bias: Bias acts as an additional free variable that can be useful for learning.

Conventional fully connected NN

- A conventional Fully Connected Neural Networks receive an input (a single vector)
- Transform the input through a series of *hidden layers*
- Each hidden layer is made up of a set of neurons, where each neuron is fully connected to all neurons in the previous layer
- Last fully-connected layer is called the “output layer” and in classification settings it represents the class scores
- There can be any number of hidden layers within a network.



Basic process of Learning an NN

- Forward pass
  - Input is passed through the network layers and the output is calculated based on network weights
- Loss function
  - Loss is calculated by comparing the desired output with the output of the network.

$$loss = (Desired\ output — network\ output)$$

- Back propagation
  - The network weights are updated based on back propagation algorithm to minimize the loss

https://mattmazur.com/2015/03/17/a-step-by-step-backpropagation-example/

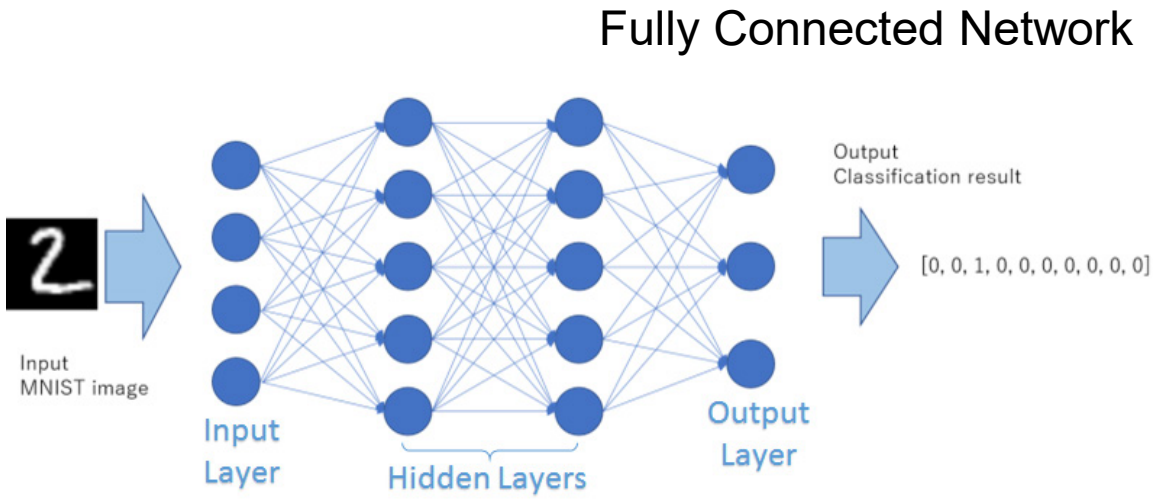
MNIST Database

- Database of handwritten digits
- Training set consist of 60,000 examples
- Test set consist of 10,000 examples

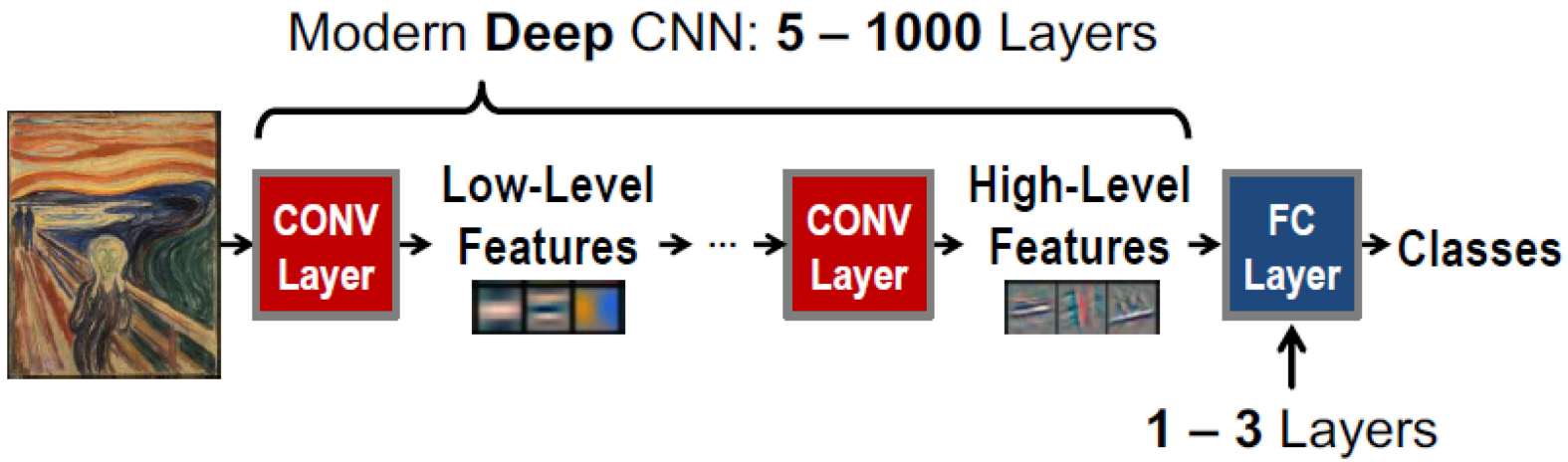


MNIST classification

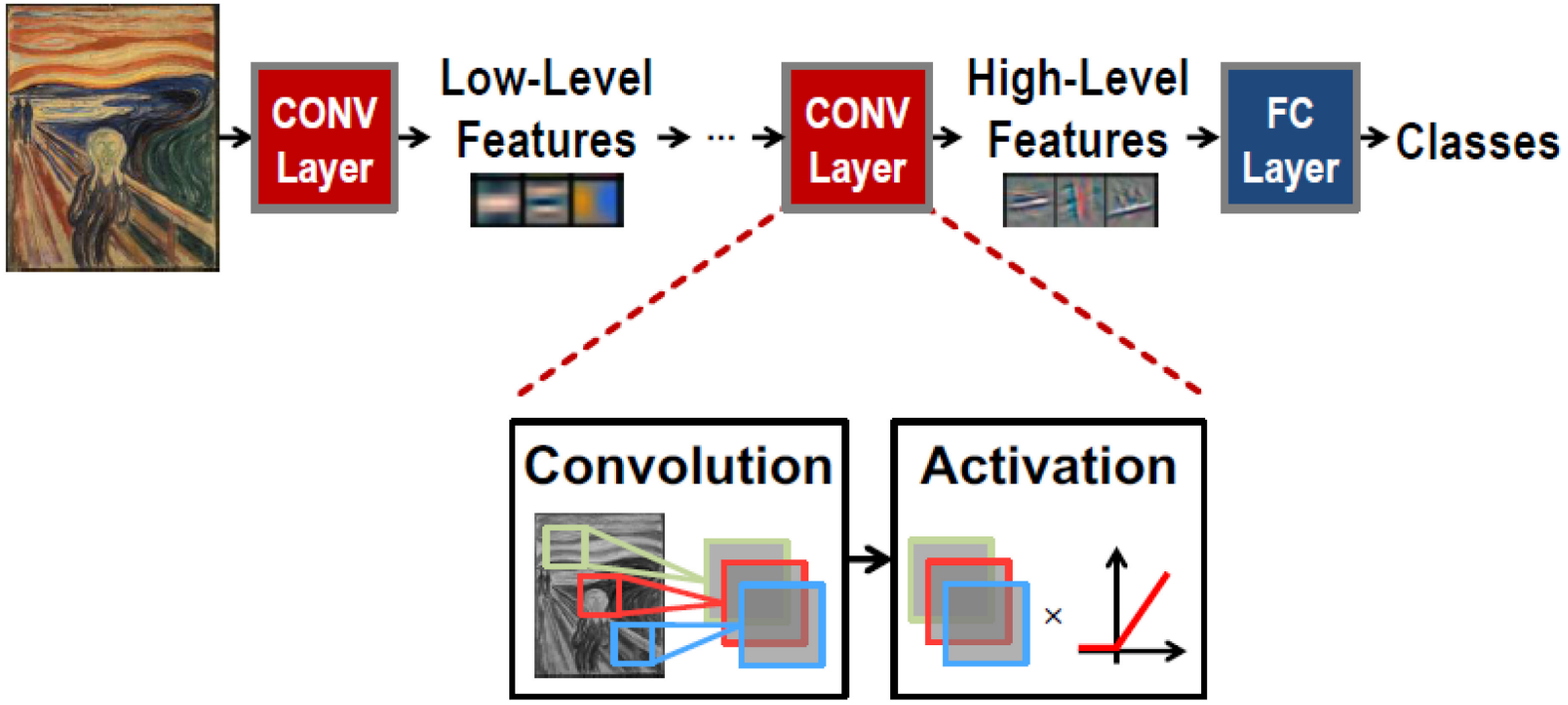
- Steps for NN architecture in Python/Keras
  1. Import Libraries
  2. Load MNIST dataset
  3. Preprocess data
  4. Define model
  5. Compile model
  6. Train the model
  7. Evaluate model



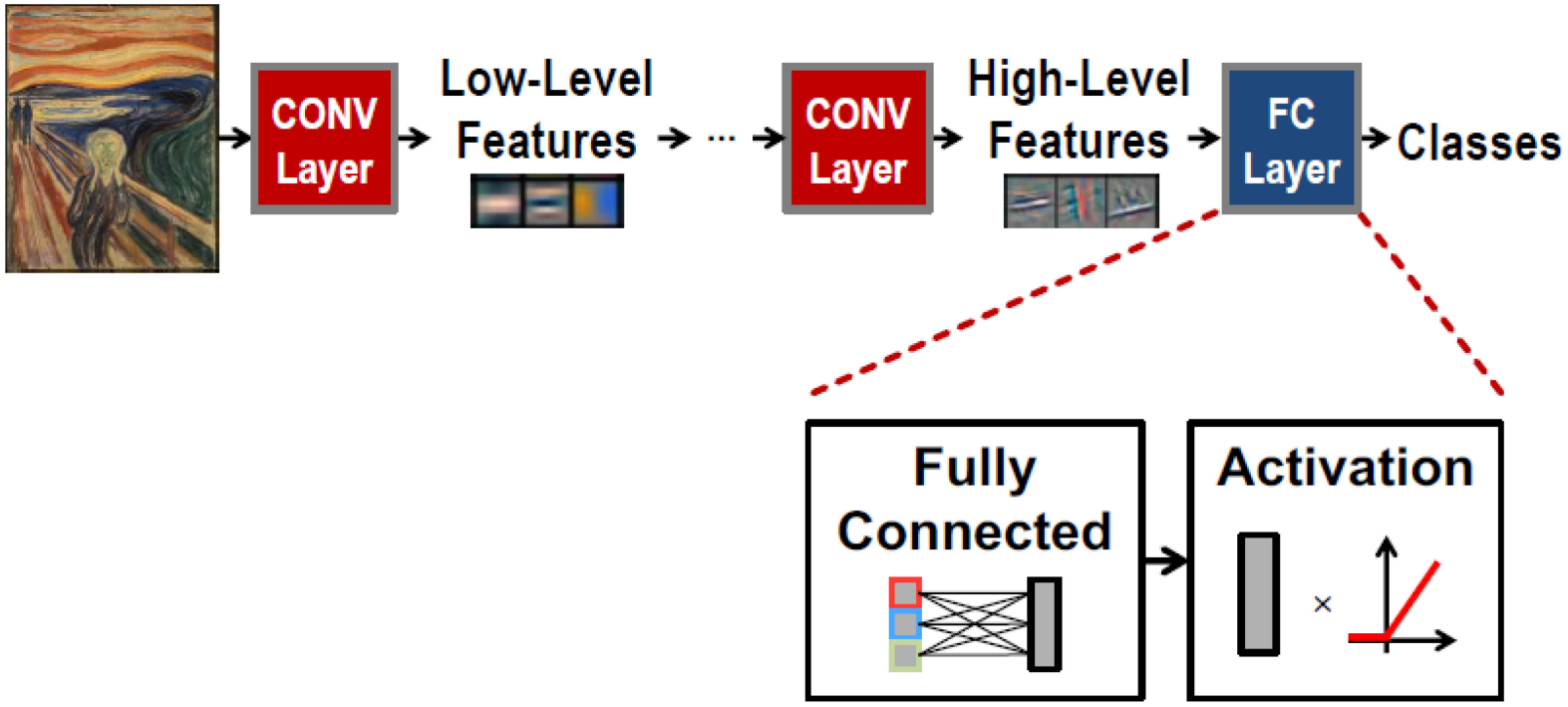
Deep Convolutional Neural Networks



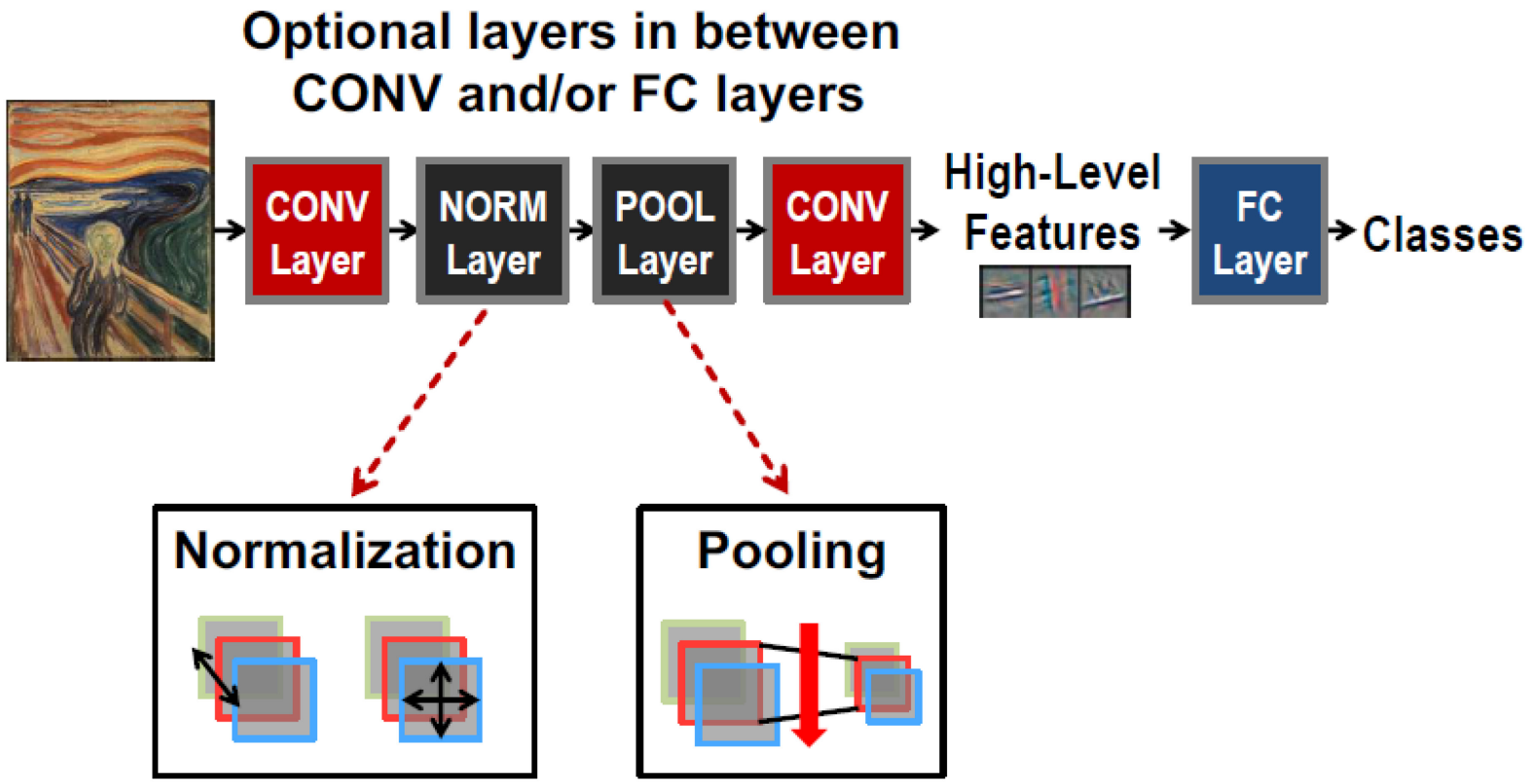
Deep Convolutional Neural Networks



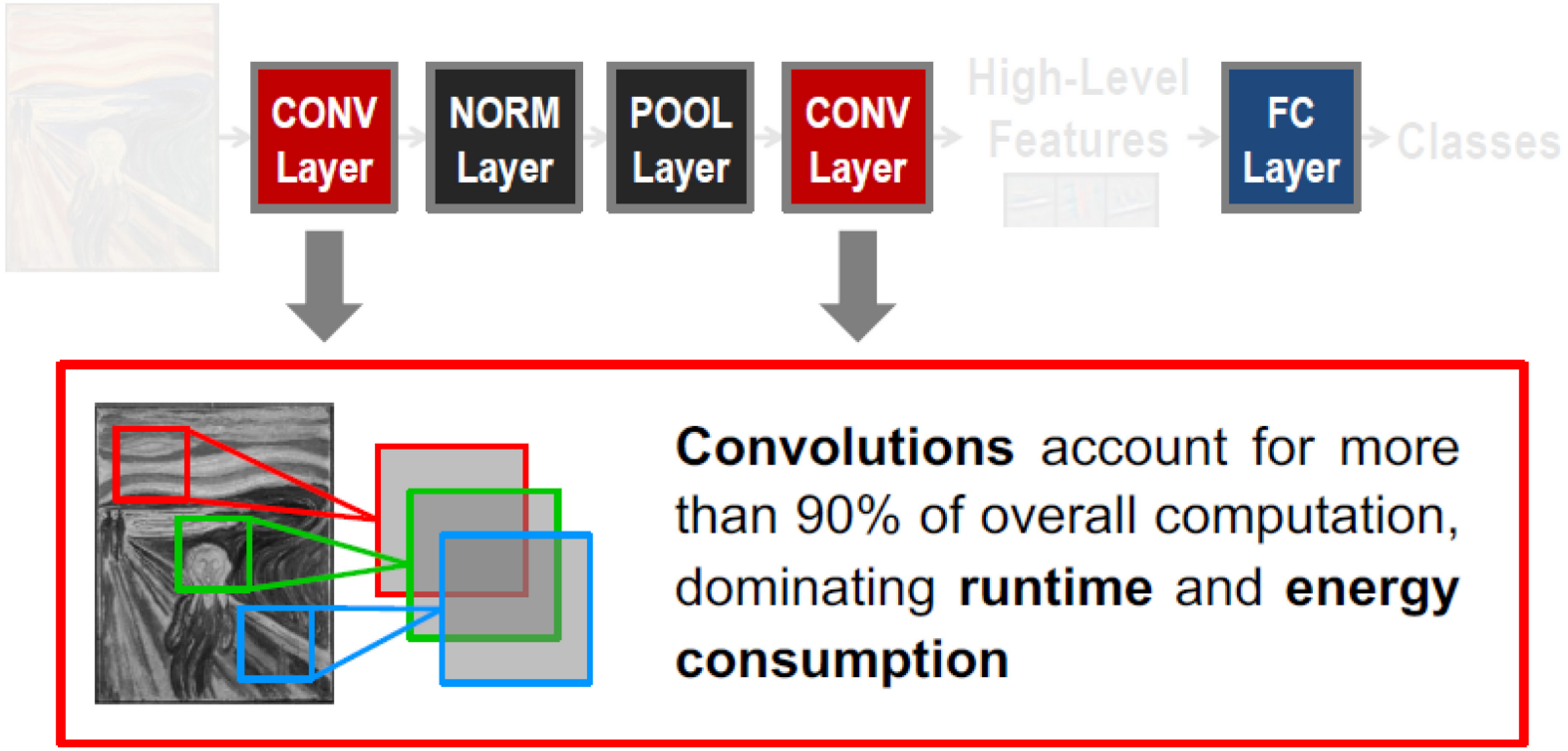
Deep Convolutional Neural Networks



Deep Convolutional Neural Networks



Deep Convolutional Neural Networks



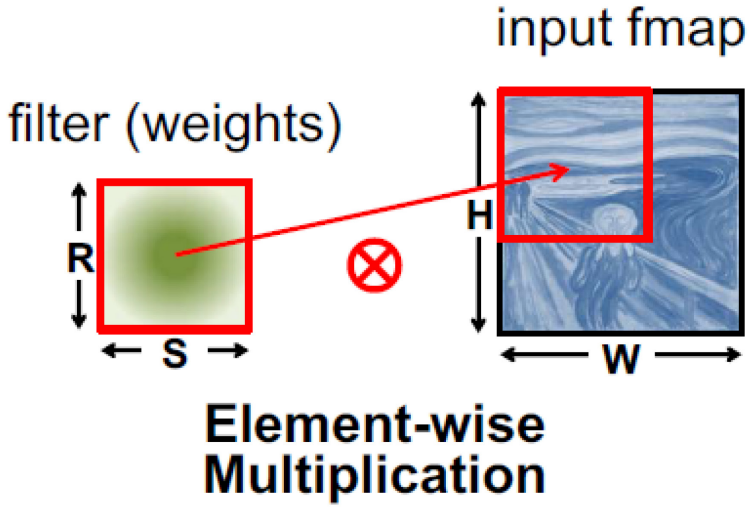
Convolution (CONV) Layer

a plane of input activations
a.k.a. **input feature map (fmap)**

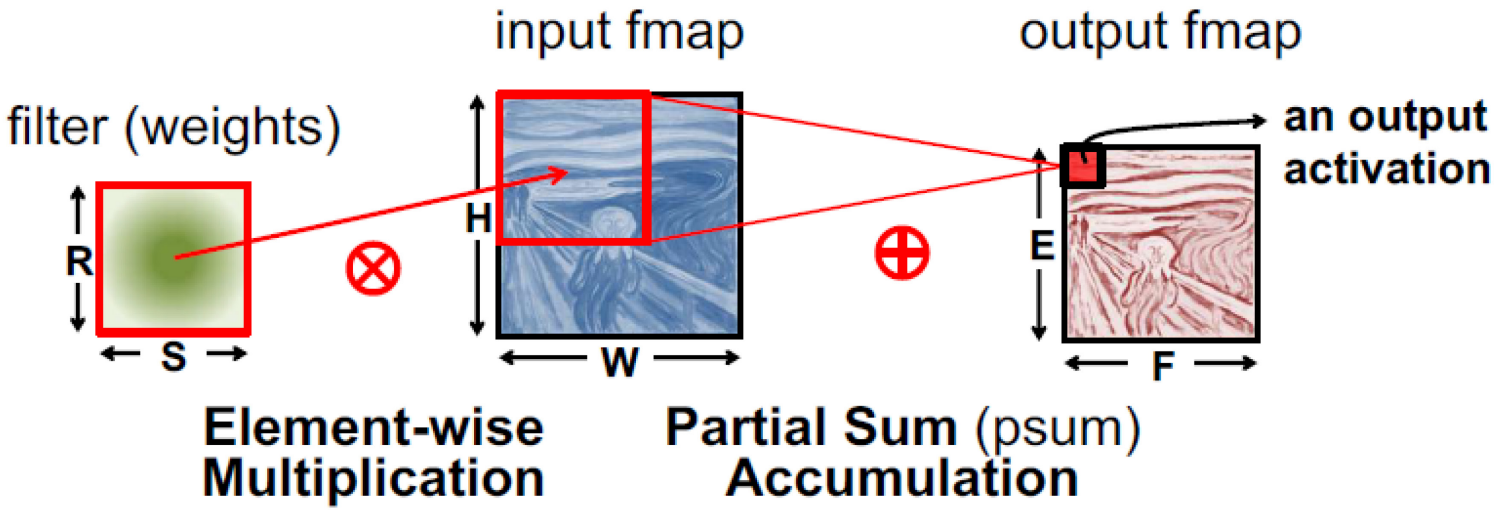
filter (weights)



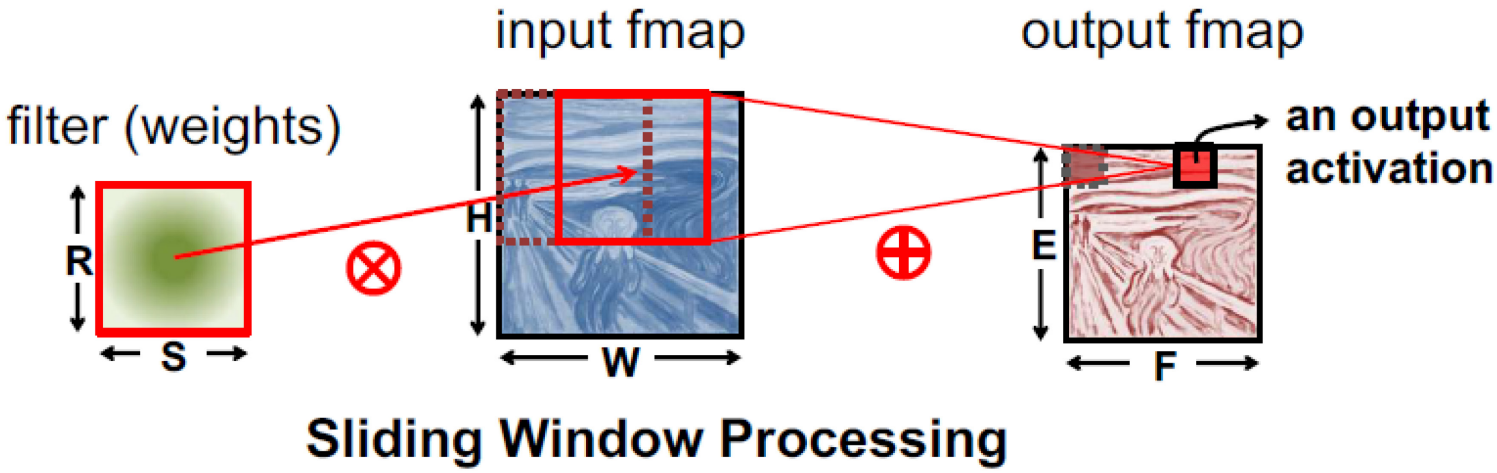
Convolution (CONV) Layer



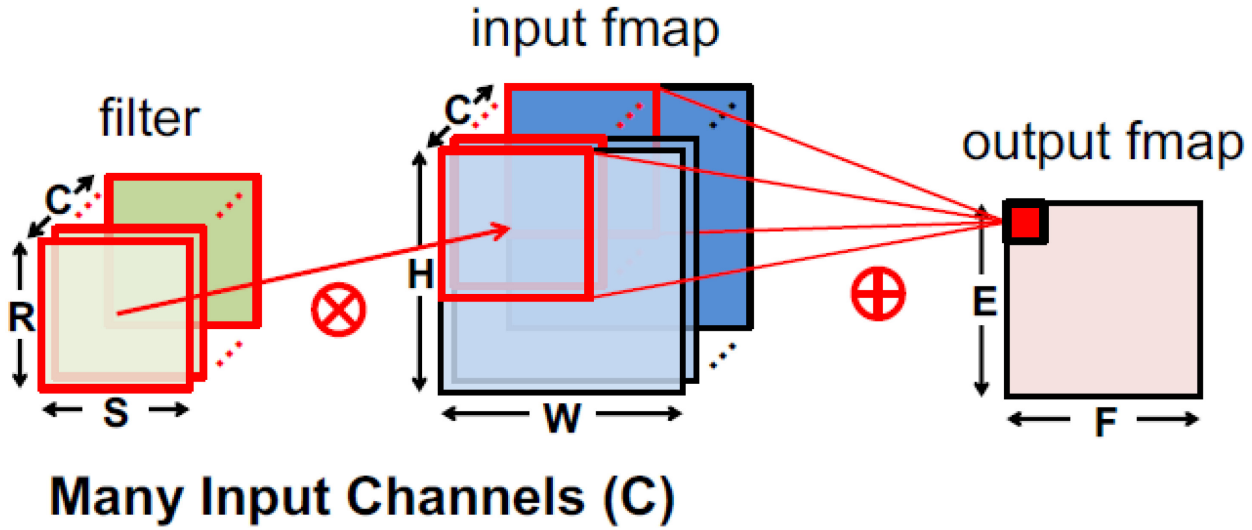
Convolution (CONV) Layer



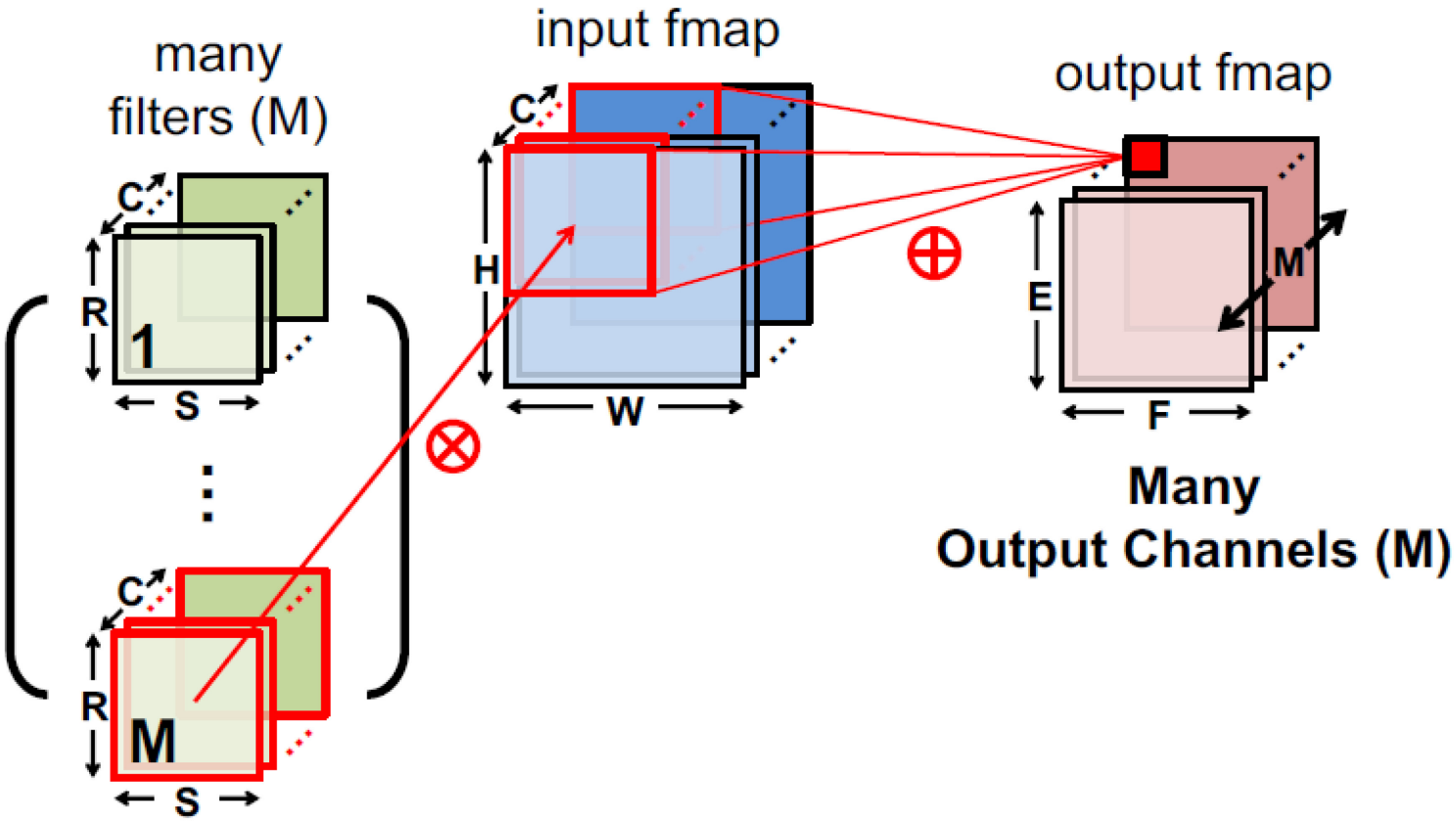
Convolution (CONV) Layer



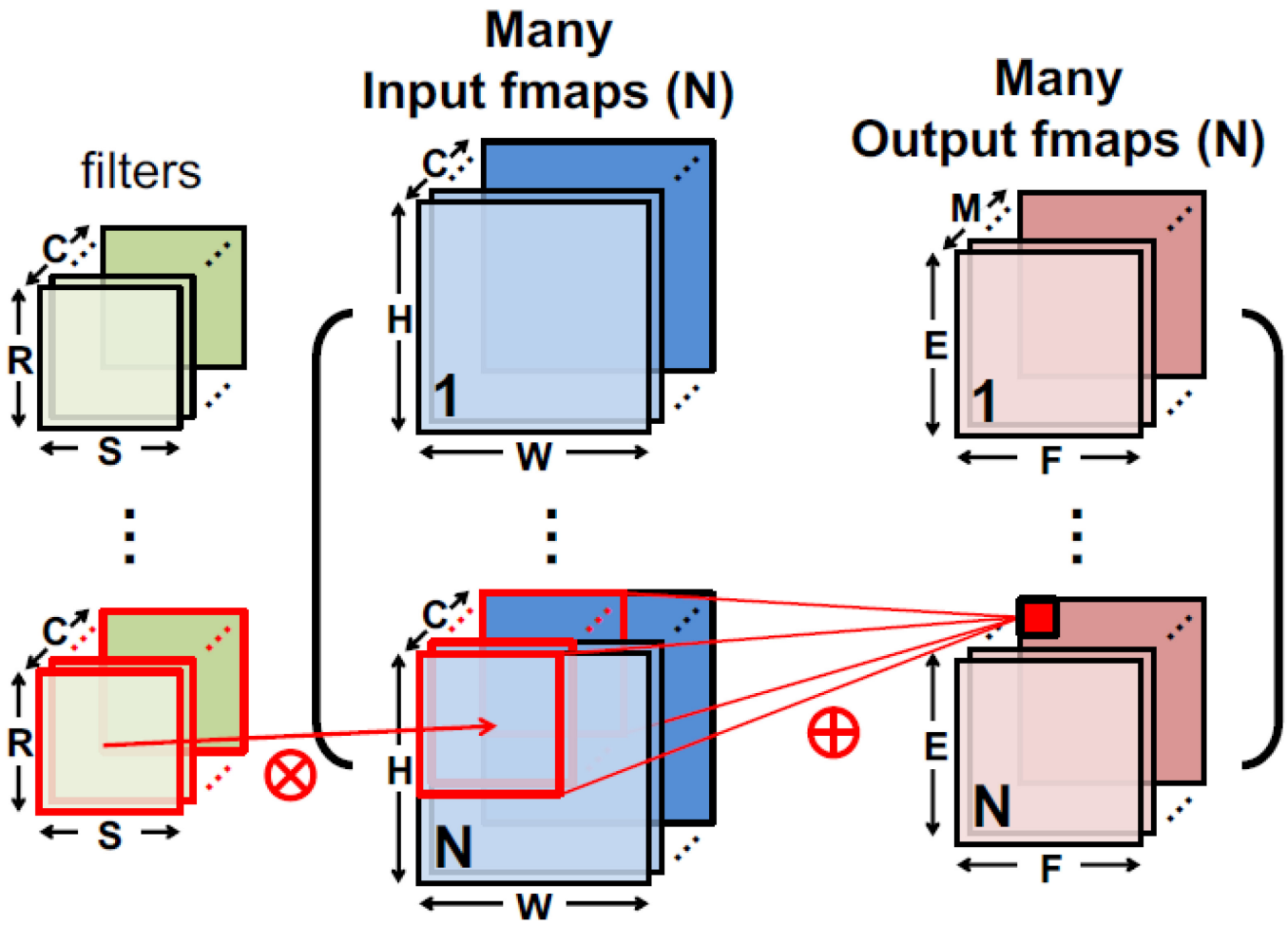
Convolution (CONV) Layer



Convolution (CONV) Layer

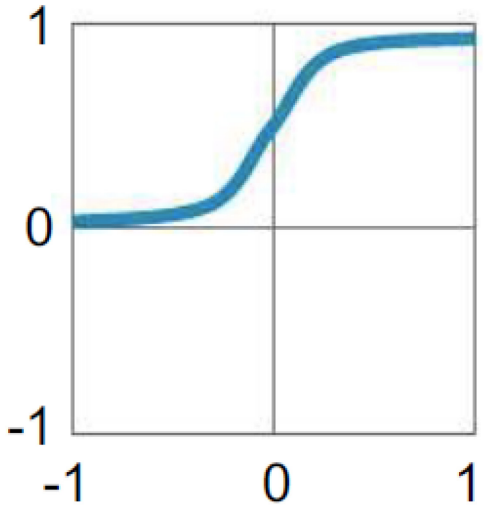


Convolution (CONV) Layer



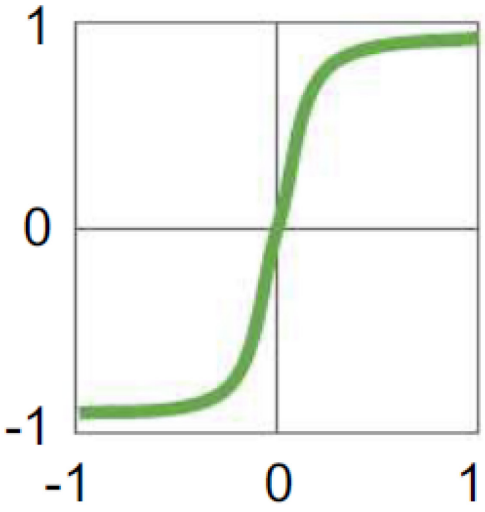
Traditional Activation Functions

Sigmoid



$$y=1/(1+e^{-x})$$

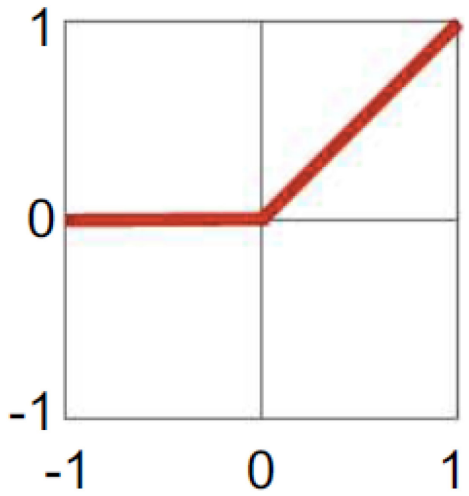
Hyperbolic Tangent



$$y=(e^{x}-e^{-x})/(e^{x}+e^{-x})$$

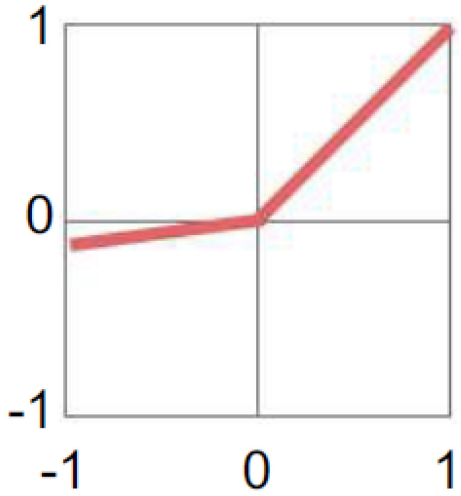
Modern Activation Functions

Rectified Linear Unit (ReLU)



$y=\max(0,x)$

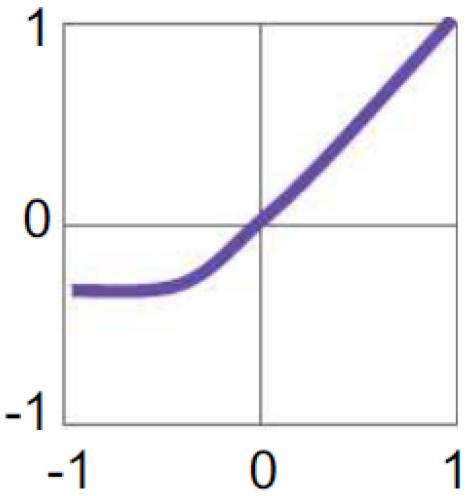
Leaky ReLU



$y=\max(\alpha x,x)$

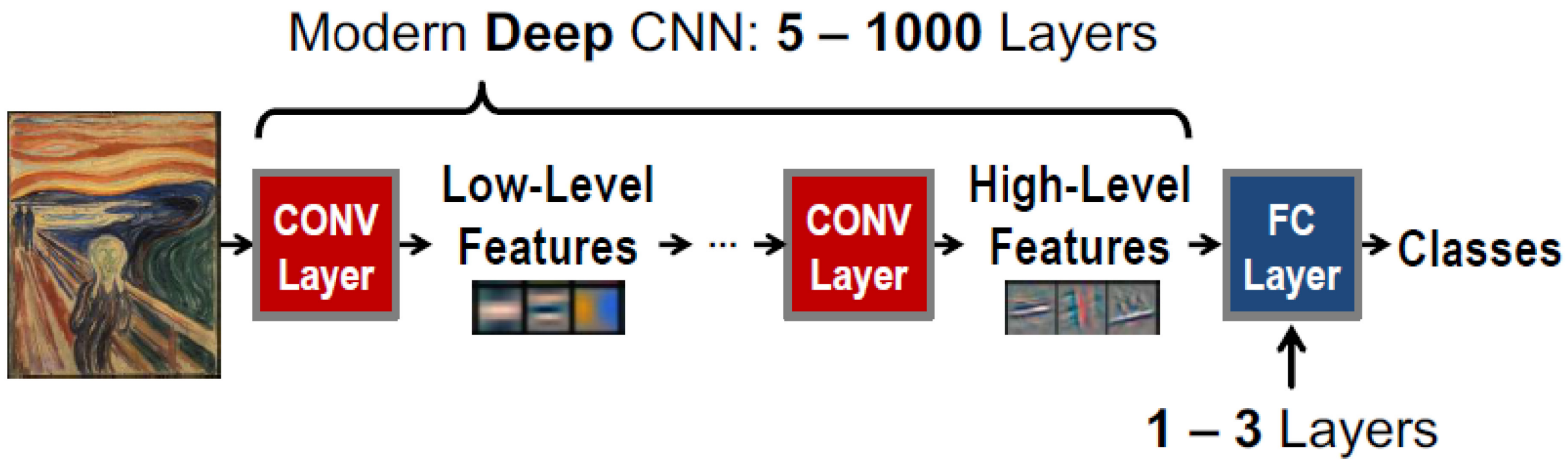
α = small const. (e.g. 0.1)

Exponential LU



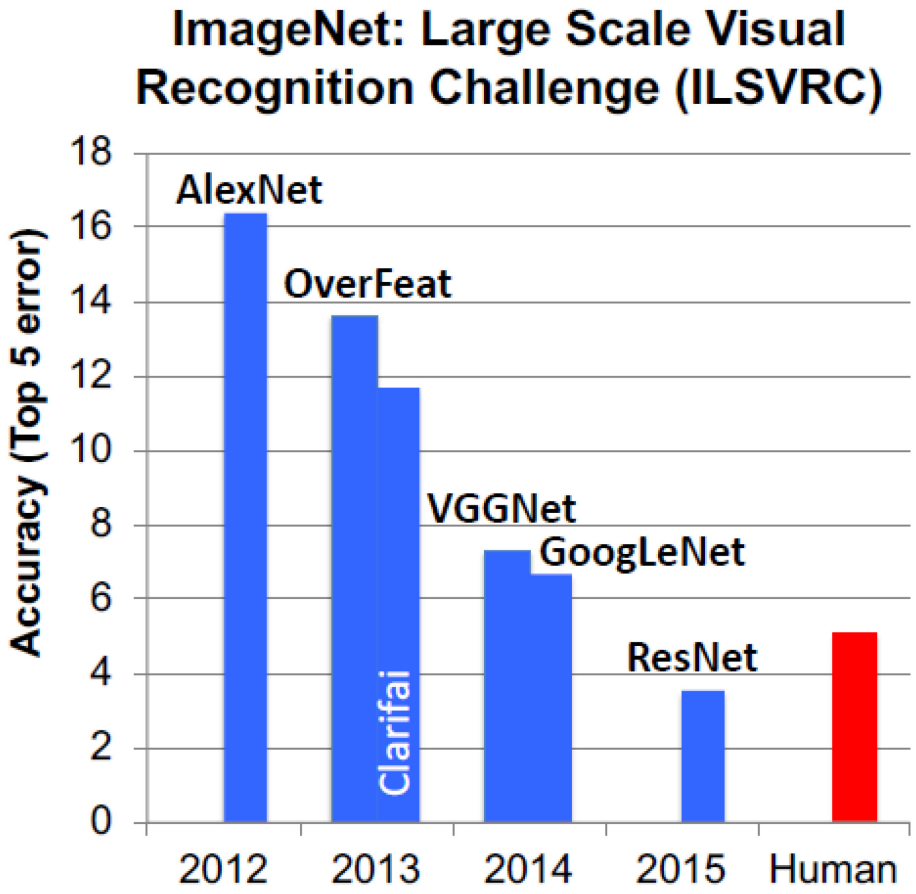
$$y=\begin{cases} x, & x\geq 0 \\ \alpha(e^{x}-1), & x<0 \end{cases}$$

Deep Convolutional Neural Networks



Popular DNN Models

- LeNet (1998)
- AlexNet (2012)
- OverFeat (2013)
- VGGNet (2014)
- GoogleNet (2014)
- ResNet (2015)



Russakovsky et al. IJCV 2015

ImageNet

IMAGENET

Image Classification
~256x256 pixels (color)
1000 Classes
1.3M Training
100,000 Testing (50,000 Validation)

For **ImageNet Large Scale Visual Recognition Challenge (ILSVRC)**
accuracy of classification task reported based on top-1 and top-5 error

Image Source: http://karpathy.github.io/



http://www.image-net.org/challenges/LSVRC/

Popular DNN Models

CONV Layers increasingly important!

Metrics	LeNet-5	AlexNet	VGG-16	GoogLeNet (v1)	ResNet-50
Top-5 error	n/a	16.4	7.4	6.7	5.3
Input Size	28x28	227x227	224x224	224x224	224x224
# of CONV Layers	**2**	**5**	**16**	**21 (depth)**	**49**
Filter Sizes	5	3, 5,11	3	1, 3 , 5, 7	1, 3, 7
# of Channels	1, 6	3 - 256	3 - 512	3 - 1024	3 - 2048
# of Filters	6, 16	96 - 384	64 - 512	64 - 384	64 - 2048
Stride	1	1, 4	1	1, 2	1, 2
# of Weights	2.6k	2.3M	14.7M	6.0M	23.5M
# of MACs	283k	666M	15.3G	1.43G	3.86G
# of FC layers	**2**	**3**	**3**	**1**	**1**
# of Weights	58k	58.6M	124M	1M	2M
# of MACs	58k	58.6M	124M	1M	2M
Total Weights	**60k**	**61M**	**138M**	**7M**	**25.5M**
Total MACs	**341k**	**724M**	**15.5G**	**1.43G**	**3.9G**

Popular DNN Models

AlexNet

- First CNN Winner of ILSVRC
- Uses LRN (deprecated after this)

VGG-16

- Goes Deeper (16+ layers)
- Uses only 3x3 filters (stack for larger filters)

GoogLeNet (v1)

- Reduces weights with Inception and only one FC layer
- Inception: 1x1 and DAG (parallel connections)
- Batch Normalization

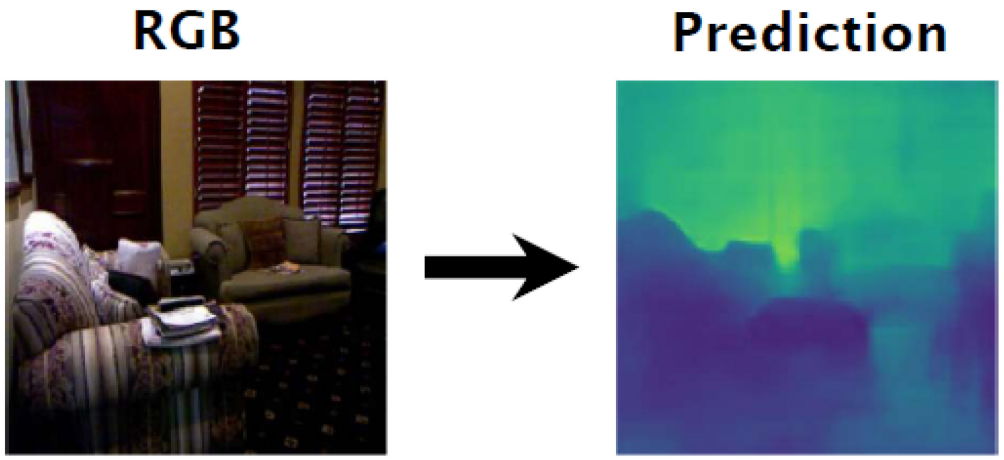
ResNet

- Goes Deeper (24+ layers)
- Shortcut connections

Applications –
Beyond Image Classification

FastDepth: Fast Monocular Depth Estimation

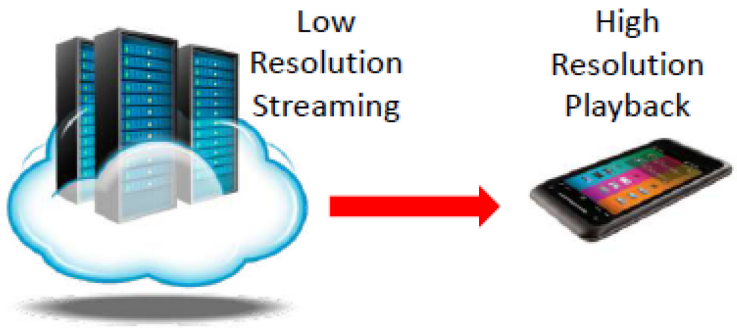
- Real-time low-power depth sensing is critical for navigation of small robotic vehicles.
- Depth estimation from a single RGB image desirable, due to the relatively low cost and size of monocular cameras.



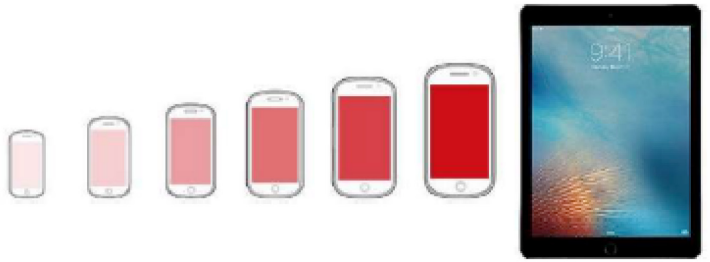
Our goal is to enable **high accuracy, low latency, high throughput monocular depth estimation** on a **deployable embedded system**.

Wofk* et al. ICRA 2019

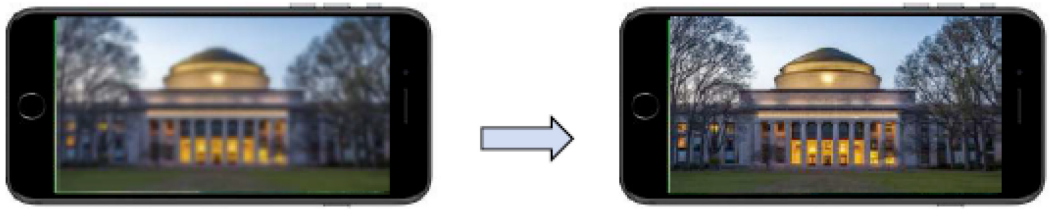
Super-Resolution CNN



Transmit low resolution for lower bandwidth



Screens are getting larger



Use **super-resolution** to improve the viewing experience of lower-resolution content (*reduce communication bandwidth*)